**Code No: 117CD** **R13**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech IV Year I Semester Examinations, July - 2021**
**DATA WAREHOUSING AND DATA MINING**
**(Computer Science and Engineering)**

**Time: 3 Hours** **Max. Marks: 75**

**Answer any Five Questions**
**All Questions Carry Equal Marks**
- - -

1.a) Explain the Data warehouse architecture and its components
b) Explain star schema with an example. [7+8]

2.a) Explain the OLAP operations in the multi dimensional model.
b) Briefly outline how to compute the dissimilarity between objects described by the following:
i) Nominal attributes
ii) Asymmetric binary attributes
iii) Numeric attributes
iv) Term-frequency vectors. [7+8]

3.a) State the need of Data Cleaning .What are the issues that have to be addressed during Data integration. Why Data transformation is essential in the process of Knowledge Discovery.
b) Describe the issues and challenges of Data Mining Systems. [7+8]

4. Define Association Rule miming. Explain FP-Growth algorithm with suitable illustration. Explain the pros and cons of Apriori algorithm. [15]

5. Apply a Naive -Bayes classification for the given training tuples in the table below. [15]

| Customer id | Gender | Car type | Shirt size | Class |
|---|---|---|---|---|
| 1 | M | FAMILY | SMALL | C0 |
| 2 | M | SPORTS | LARGE | C0 |
| 3 | M | SPORTS | EXTRA LARGE | C0 |
| 4 | M | SPORTS | EXTRA LARGE | C0 |
| 5 | M | FAMILY | LARGE | C0 |
| 6 | M | FAMILY | EXTRA LARGE | C1 |
| 7 | M | FAMILY | MEDIUM | C1 |
| 8 | F | LUXURY | MEDIUM | C1 |
| 9 | F | LUXURY | MEDIUM | C1 |
| 10 | F | LUXURY | MEDIUM | C1 |

JNTUH USED Paper-2021

6.a) Write the basic algorithm for inducing a decision tree from training tuples. And also explain it.

b) What is "Over FIT"? Explain about the disadvantages of an over fitted DE-CISION TREE. With an example explain the use of Tree -Pruning. [7+8]

7.a) Explain PAM algorithms.

b) Explain K-means algorithm. [7+8]

8. Discuss various Hierarchical clustering methods. (Agglomerative, divisive, Chameleon and BIRCH). [15]

**---ooOoo---**

JNTUH USED Paper-2021